RIS-aided Scalable Electromagnetic Fluid Vector Antenna MIMO Radar for Target Localization

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ABSTRACT

This paper proposes a reconfigurable intelligent surface (RIS)-aided scalable electromagnetic fluid vector antenna (SEFVA) multiple-input multiple-output (MIMO) radar system for non-line-of-sight (NLOS) far-field target localization. First, a bistatic SEFVA MIMO radar model assisted by RIS is established, taking into account the mutual coupling effects under the compressed array configuration. Subsequently, by applying parallel factor (PARAFAC) decomposition, estimation of signal parameters via rotational invariance techniques (ESPRIT), and normalized vector cross product (VCP), automatic pairing two-dimensional (2-D) direction-of-departure (DOD), 2-D direction-of-arrival (DOA), and polarization parameters for multiple targets are achieved, among which spatial smoothing technique is applied to effectively suppress the edge effects of mutual coupling in the compressed array configuration. Simulation results validate the effectiveness of the proposed algorithm.

***Index Terms*—** SFA, EMVS, MIMO, DOA, RIS, mutual coupling.

1. INTRODUCTION

As an advanced array radar system, the electromagnetic vector sensor multiple-input multiple-output (EMVS-MIMO) radar can simultaneously acquire the spatial orientation of targets, including two-dimensional (2-D) direction-of-departure (DOD) and 2-D direction-of-arrival (DOA), as well as full polarization information. It demonstrates significant advantages in fields such as target detection, recognition, and tracking [1–4]. Compared to traditional scalar sensor arrays, EMVS, by virtue of its inherent multi-component structure, achieves higher spatial degrees of freedom and parameter identification capabilities even with a limited number of array elements. This makes it particularly suitable for application scenarios where platform space is constrained or system costs are sensitive [5–7].

However, traditional EMVS-MIMO radar systems based on fixed-position array (FPA) still face several fundamental challenges in practical applications. Firstly, the fixed geometric structure of the array makes it difficult to address the inherent trade-off between avoiding grating lobe ambiguity (which requires element spacing less than half a wavelength) and suppressing mutual coupling effects (which requires larger element spacing). Secondly, although sparse arrays or non-uniform arrays can expand the physical aperture and improve resolution, they often introduce severe phase ambiguity issues.

In recent years, with the emergence of new hardware technologies such as reconfigurable intelligent surface (RIS) [8–10] and fluid antenna systems (FAS) [11], EMVS-MIMO radar systems are poised to gain further development opportunities in areas such as non-line-of-sight (NLOS) localization, dynamic environment adaptation, and flexible array configuration. To this end, this paper proposes a dynamic aperture adjustment framework based on scalable fluid antennas, constructing a novel RIS-aided scalable electromagnetic fluid vector antenna (SEFVA) MIMO radar system. Through a software-controllable array aperture scaling mechanism, the system can dynamically switch between compressed and extended configurations. Considering a far-field signal propagation model, this paper first reconstructs the received array data into a parallel factor (PARAFAC) model. Then, using the trilinear alternating least squares (TALS) algorithm and vector cross product (VCP) technique, closed-form estimation and automatic pairing of 2-D DODs, 2-D DOAs and polarization parameters for multiple targets are achieved.

Notations: $(\cdot)^{-1}, (\cdot)^{T}, (\cdot)^{\dagger}, (\cdot)^{H}$ and $\|\cdot\|_F$ signify the inverse, transpose, pseudo-inverse, conjugate transpose (Hermitian), and Frobenius norm of a matrix, respectively. $\circ, \circledast, \otimes$ and $\odot$ represent the outer product, vector outer product, Kronecker product and Khatri-Rao product, respectively. $\mathbf{I}_N$ is the $N \times N$ identity matrix and its n-th row is denoted by $i_{N,n}$. $angle\{\cdot\}, diag\{\cdot\}$ stand for the phase and diagonalization, respectively.

2. SIGAL MODEL

This paper investigates a scenario involving a RIS-assisted SEFVA MIMO radar system with K far-field targets, as illustrated in Fig. 1. The Tx array and Rx array are composed of an M element arbitrary electromagnetic vector (EMV) antenna array and an N element scalable uniform rectangular

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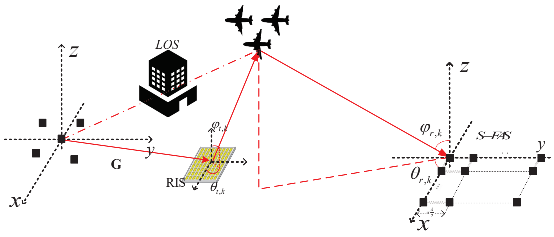


Fig. 1. The RIS-aided SEFVA MIMO radar model.

electromagnetic fluid vector (EFV) antenna array, respectively. The positions of the Rx array can be adjusted according to a scaling factor α, which modifies the inter-element spacing to be $\alpha\frac{\lambda}{2}$. The electric field signal from the transmitting array, after being reflected by the RIS and arriving at the target, can be expressed as:

$$\begin{aligned} \mathbf{r}_{m,k} &= \mathbf{V}_{t,k}^{\mathrm{T}}\mathbf{d}_m[[\mathbf{G}]_m diag\left\{\mathbf{w}\right\}\mathbf{a}_{t,k}]^{\mathrm{T}} s_m\left(t\right) \\ &= \zeta_k[[\mathbf{G}]_m diag\left\{\mathbf{w}\right\}\mathbf{a}_{t,k}]^{\mathrm{T}} s_m\left(t\right) \end{aligned} \tag{1}$$

where $\zeta_k = \mathbf{V}_{t,k}^{\mathrm{T}}\mathbf{d}_m = [\zeta_{k,H}, \zeta_{k,V}]^T \in \mathbb{C}^{2\times 1}$ represents the probing polarization of the transmitting EMV antenna array, encompassing both the H and V components of the waveform. $s_m\left(t\right)$ denotes the waveform transmitted by the m-th EMV element. $[\mathbf{G}]_m$ signifies the channel matrix from the m-th transmitting element to the RIS, which incorporates spatial channel information comprising distance attenuation and propagation phase delay, defined as:

$$[\mathbf{G}]_m = \left[\begin{array}{cccc} e^{-j2\pi\rho_{m,1}/\lambda} & e^{-j2\pi\rho_{m,2}/\lambda} & \cdots & e^{-j2\pi\rho_{m,K}/\lambda} \end{array}\right] \tag{2}$$

where $\rho_{m,q} = \left|\mathbf{p}_q^s - \mathbf{p}_m^t\right|$ denotes the distance between the q-th transmitting element and the m-th RIS reflecting element, and the transmit polarization weight vector $\mathbf{w} = \left[\alpha_1 e^{j\phi_1}, \ldots, \alpha_Q e^{j\phi_Q}\right]^T$ is used to control the polarization state of the transmitted waveform.

The steering vector from the RIS to the k-th target, denoted as $\mathbf{a}_{t,k}$, is defined as:

$$\mathbf{a}_{t,k} = \left[e^{-j2\pi\sigma_{1,k}/\lambda}, e^{-j2\pi\sigma_{2,k}/\lambda}, \cdots, e^{-j2\pi\sigma_{q,k}/\lambda}\right]^{\mathrm{T}} = e^{-j2\pi\langle\boldsymbol{\Omega}\rangle_k^{\mathrm{T}}/\lambda} \tag{3}$$

where $\sigma_{q,k} = \left(\mathbf{p}_q^s - \mathbf{p}_1^s\right)\mathbf{q}_{t,k}$, $\boldsymbol{\Omega} = \left(\mathbf{P}^s - [\mathbf{P}]_1^s\right)\mathbf{Q}_t$. $\mathbf{P}^s = \left[\left(\mathbf{p}_1^s\right)^T, \left(\mathbf{p}_2^s\right)^T, \cdots, \left(\mathbf{p}_Q^s\right)^T\right]^T$ represents the position matrix of the RIS, with $[\mathbf{P}]_i^s$ denoting the i-th row of the matrix, and $\mathbf{Q}_t = [q_{t,1}, q_{t,2}, \cdots, q_{t,K}]$ signifies the direction matrix formed by column-wise stacking.

In practical arrays, due to the electromagnetic mutual coupling effects between antenna elements, the actual steering vector deviates from the ideal model. Therefore, it is necessary to introduce a mutual coupling matrix:

$$\mathbf{C} = \begin{bmatrix} 1 & c_1 & c_2 & \cdots & c_{N-1} \\ c_1^* & 1 & c_1 & \cdots & c_{N-2} \\ c_2^* & c_1^* & 1 & \cdots & c_{N-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_{N-1}^* & c_{N-2}^* & c_{N-3}^* & \cdots & 1 \end{bmatrix} \tag{4}$$

where the mutual coupling coefficient $c_n(\alpha)$ is defined as:

$$c_n(\alpha) = c_0 \exp\left(-\beta\frac{n\cdot\alpha\cdot\frac{\lambda}{2}}{\lambda}\right) e^{j\psi_n(\alpha)} \tag{5}$$

where the phase component is given by

$$\psi_n(\alpha) = \frac{2\pi m\cdot\alpha\cdot\frac{\lambda}{2}}{\lambda} + \phi_0, \tag{6}$$

c_0 denotes the reference coupling strength at the baseline spacing, β is the coupling attenuation factor, and ϕ_0 represents the phase offset.

The steering vector is defined as:

$$\mathbf{a}_k = \left[\frac{r_k}{r_{1,k}} e^{j\frac{2\pi}{\lambda}(r_{1,k}-r_k)}, \frac{r_k}{r_{2,k}} e^{j\frac{2\pi}{\lambda}(r_{2,k}-r_k)}, \cdots, \frac{r_k}{r_{N,k}} e^{j\frac{2\pi}{\lambda}(r_{N,k}-r_k)}\right]^{\mathrm{T}} \tag{7}$$

where $r_{n,k}(\alpha) = \sqrt{{r_k}^2 + p_n^2(\alpha) - 2r_k p_n(\alpha)\cdot(\sin\theta_k\cos\phi_k)}$, the position of the n-th antenna element is given by $p_n(\alpha) = (n-1)\cdot\alpha\cdot\frac{\lambda}{2}$, with α being the scaling factor (compressed configuration: $\alpha_c < 1$ or extended configuration: $\alpha_e > 1$. The system operates by intelligently switching between these two complementary working modes: the compressed configuration mitigates grating lobe ambiguity at the cost of enhanced mutual coupling effects, whereas the extended configuration significantly improves spatial resolution while rendering mutual coupling effects negligible. This paper only considers scenarios where the array aperture is sufficiently small relative to the source distance, allowing the spherical wavefront in the model to be approximated as a plane wave. Under this condition, the exact distance is $\mathbf{p}_n^r\mathbf{q}_{r,k}$, and the steering vector at the receiver is denoted as:

$$\mathbf{a}_{r,k} = \left[e^{-j2\pi\mathbf{p}_n^1\mathbf{q}_{1,k}/\lambda}, e^{-j2\pi\mathbf{p}_n^2\mathbf{q}_{2,k}/\lambda}, \cdots, e^{-j2\pi\mathbf{p}_n^r\mathbf{q}_{r,k}/\lambda}\right]^{\mathrm{T}} = e^{-j2\pi\mathbf{P}^r\mathbf{Q}_r/\lambda} \tag{8}$$

where $\mathbf{P}^r = \left[\left(\mathbf{p}_1^r\right)^T, \left(\mathbf{p}_2^r\right)^T, \cdots, \left(\mathbf{p}_N^r\right)^T\right]^T$ represents the position matrix of the receiving EFV elements. $\mathbf{Q}_r = [\mathbf{q}_{r,1}, \mathbf{q}_{r,2}, \cdots, \mathbf{q}_{r,K}]$ denotes the direction matrix formed by column-wise stacking.The receiver-side steering vector accounting for mutual coupling is denoted as $\mathbf{h}_{r,k}(\alpha)$.

After matched filtering, the outputs at the receiving SEFVA array can be expressed as

$$\begin{aligned} \mathbf{y}(\tau) &= \sum_{k=1}^{K}\left([\mathbf{G}diag\left\{\mathbf{w}\right\}\mathbf{a}_{t,k}]\otimes\mathbf{h}_{r,k}(\alpha)\otimes\mathbf{b}_k\right)u_k\left(\tau\right) + \mathbf{n}\left(\tau\right) \\ &= [\mathbf{G}diag\left\{\mathbf{w}\right\}\mathbf{A}_t\odot\mathbf{H}_r\odot\mathbf{B}]\,\mathbf{u}\left(\tau\right) + \mathbf{n}(\tau) \end{aligned} \tag{9}$$

where $\mathbf{A}_t=[\mathbf{a}_{t,1},\mathbf{a}_{t,2},\cdots,\mathbf{a}_{t,k}]\in\mathbb{C}^{Q\times K},\mathbf{B}=[\mathbf{b}_1,\mathbf{b}_2,\cdots,\mathbf{b}_K]\in\mathbb{C}^{6\times K}$, $\mathbf{H}_r=[\mathbf{h}_{r,1},\mathbf{h}_{r,2},\cdots,\mathbf{h}_{r,K}]\in\mathbb{C}^{N\times K}$, $\mathbf{u}(\tau)=[u_1(\tau),u_2(\tau),\cdots,u_K(\tau)]^{\mathrm{T}}\in\mathbb{C}^{K\times 1}$ and $\mathbf{n}(\tau)=\left[\mathbf{n}_1^{\mathrm{T}}(\tau),\cdots,\mathbf{n}_m^{\mathrm{T}}(\tau)\right]^{\mathrm{T}}\in\mathbb{C}^{6MN\times 1}$. The model for $\mathbf{b}$ can be expressed as:

$$\mathbf{b}=\mathbf{V}\mathbf{g} \tag{10}$$

$$\mathbf{V}=\begin{bmatrix}\cos(\theta)\cos(\varphi) & -\sin(\theta)\\ \sin(\theta)\cos(\varphi) & \cos(\theta)\\ -\sin(\varphi) & 0\\ -\sin(\theta) & -\cos(\theta)\cos(\varphi)\\ \cos(\theta) & -\sin(\theta)\cos(\varphi)\\ 0 & \sin(\varphi)\end{bmatrix},\mathbf{g}=\begin{bmatrix}\sin(\gamma)e^{j\eta}\\ \cos(\gamma)\end{bmatrix} \tag{11}$$

where $\theta\in[-\pi,\pi)$, $\varphi\in[0,\pi]$, $\gamma\in[0,\pi/2)$ and $\eta\in[-\pi,\pi)$ represent the azimuth angle, elevation angle, auxiliary polarization angle, and polarization phase difference, respectively.

After straightforward rearrangement, the signal vector can be obtained as:

$$\mathbf{y}(\tau)=[\mathbf{G}diag\{\mathbf{w}\}\mathbf{A}_t\odot\mathbf{H}_r\odot\mathbf{B}]\mathbf{u}(\tau)+\mathbf{n}(\tau). \tag{12}$$

Expressed in matrix form, it becomes

$$\begin{aligned}\mathbf{Y}&=[\mathbf{G}diag\{\mathbf{w}\}\mathbf{A_t}\odot\mathbf{H}_r\odot\mathbf{B}]\mathbf{U}+\mathbf{N}\\&=[\mathbf{V_t}\odot\mathbf{H}_r\odot\mathbf{B}]\mathbf{U}+\mathbf{N}\end{aligned} \tag{13}$$

where $\mathbf{V_t}=\mathbf{G}diag\{\mathbf{w}\}\mathbf{A_t}\in\mathbb{C}^{M\times K}$, $\mathbf{Y}=[\mathbf{y}(1),\mathbf{y}(2),\cdots,\mathbf{y}(L)]\in\mathbb{C}^{6MN\times L}$, $\mathbf{U}=[\mathbf{u}(1),\mathbf{u}(2),\ \cdots,\mathbf{u}(L)]\in\mathbb{C}^{K\times L}$ and $\mathbf{N}=[\mathbf{n}(1),\mathbf{n}(2),\cdots,\mathbf{n}(L)]\in\mathbb{C}^{6MN\times L}$.

3. PARAMETER ESTIMATION

3.1. 2-D DOD Estimation

This paper primarily focuses on the third-order PARAFAC model obtained after the rearrangement and compression of matrix $\mathbf{Y}$ [2], which can be further expressed as:

$$\boldsymbol{\mathcal{Z}}=\sum_{k=1}^{K}\mathbf{g}_{1,k}\circ\mathbf{g}_{2,k}\circ\mathbf{g}_{3,k}+\boldsymbol{\mathcal{N}}. \tag{14}$$

Let $\mathbf{g}_{1,k}=\mathbf{v}_{t,k}$, $\mathbf{g}_{2,k}=\mathbf{h}_{r,k}\circ\mathbf{b}_{r,k}$, $\mathbf{g}_{3,k}=\mathbf{u}_k$. TALS [2] is employed to find estimates of the factor matrices $\mathbf{G}_1,\mathbf{G}_2,\mathbf{G}_3$.

By eliminating scale and phase ambiguities, we obtain

$$\hat{\mathbf{A}}_t=\left\{(\mathbf{G}diag\{\mathbf{w}\})^{\mathrm{H}}(\mathbf{G}diag\{\mathbf{w}\})\right\}^{\dagger}(\mathbf{G}diag\{\mathbf{w}\})^{\mathrm{H}}\hat{\mathbf{G}}_1. \tag{15}$$

The estimated matrix containing the 2-D DOD can be computed as:

$$\hat{\boldsymbol{\Omega}}=\frac{angle(\hat{\mathbf{A}}_t)\lambda}{2\pi}. \tag{16}$$

Thus, we have

$$\hat{\mathbf{Q}}_t=\left[(\mathbf{P}^s-\mathbf{P}_1^s)^{\mathrm{H}}(\mathbf{P}^s-\mathbf{P}_1^s)\right]^{\dagger}(\mathbf{P}^s-\mathbf{P}_1^s)^{\mathrm{H}}\hat{\boldsymbol{\Omega}}. \tag{17}$$

Finally, the 2-D DOD estimates are obtained via the following formula:

$$\begin{aligned}\hat{\theta}_{t,k}&=\arctan\left(\hat{\mathbf{Q}}_t(2,k)/\hat{\mathbf{Q}}_t(1,k)\right)\\ \hat{\varphi}_{t,k}&=\arcsin\left(\sqrt{\hat{\mathbf{Q}}_t(1,k)^2+\hat{\mathbf{Q}}_t(2,k)^2}\right).\end{aligned} \tag{18}$$

3.2. 2-D DOA Estimation

Clearly, the polarization response vector $\mathbf{b}$ can be written in normalized form as:

$$b_{r,k}=b_{r,k}(1)\left[\underbrace{1,\frac{b_{r,k}(2)}{b_{r,k}(1)},\frac{b_{r,k}(3)}{b_{r,k}(1)}}_{\mathbf{e}_{r,k}},\underbrace{\frac{b_{r,k}(4)}{b_{r,k}(1)},\frac{b_{r,k}(5)}{b_{r,k}(1)},\frac{b_{r,k}(6)}{b_{r,k}(1)}}_{\mathbf{m}_{r,k}}\right]. \tag{19}$$

To mitigate the strong edge effects of mutual coupling while maintaining computational efficiency [12], spatial smoothing is applied by selecting the central subarray. This is achieved through a selection matrix $\mathbf{F}\in\mathbb{R}^{(N-2P)\times N}$, which extracts the central $N-2P$ elements as follows:

$$\mathbf{F}=[\mathbf{0}_{(N-2P)\times P},\mathbf{I}_{N-2P},\mathbf{0}_{(N-2P)\times P}]. \tag{20}$$

For the receiving array, its spatial response matrix satisfies rotational invariance such that

$$\mathbf{H}_p\mathbf{G}_2=\mathbf{H}_1\mathbf{G}_2\boldsymbol{\Theta}_{r,p}. \tag{21}$$

The diagonal elements of $\boldsymbol{\Theta}_{r,p}=diag\left\{\frac{b_{r,1}(p)}{b_{r,1}(1)},\frac{b_{r,2}(p)}{b_{r,2}(1)},\cdots,\frac{b_{r,K}(p)}{b_{r,K}(1)}\right\}$ reflect the phase delay between subarray p and the reference subarray for the k-th source, which is directly related to the direction cosines of the source. $\mathbf{H}_p$ is a selection matrix satisfying $\mathbf{H}_p=\mathbf{I}_{N-2P}\otimes\mathbf{i}_{6,p}$. Based on the reduced-dimensional factor matrix $\hat{\mathbf{G}}_2'=(\mathbf{F}\otimes\mathbf{I}_6)\hat{\mathbf{G}}_2$, the estimated value of the phase difference matrix can be solved via the least squares criterion. Taking into account the influence of the ambiguity matrix $\boldsymbol{\Pi}$, the estimation formula is

$$\hat{\boldsymbol{\Theta}}_{r,p}=\hat{\boldsymbol{\Pi}}(\mathbf{H}_1\hat{\mathbf{G}}_2')^{\dagger}\mathbf{H}_p\hat{\mathbf{G}}_2'\hat{\boldsymbol{\Pi}}^{-1} \tag{22}$$

From this, the estimated Poynting vector can be obtained [2]

$$[\hat{u}_{r,k},\hat{v}_{r,k},\hat{w}_{r,k}]^T=\frac{\hat{\mathbf{e}}_{r,k}}{\|\hat{\mathbf{e}}_{r,k}\|_F}\circledast\frac{\hat{\mathbf{m}}_{r,k}^*}{\|\hat{\mathbf{m}}_{r,k}\|_F}. \tag{23}$$

Finally, the 2-D DOA estimates are obtained as:

$$\begin{aligned}\hat{\theta}_{r,k}&=\arctan\left(\hat{v}_{r,k}/\hat{u}_{r,k}\right)\\ \hat{\varphi}_{r,k}&=\arcsin\left(\sqrt{\hat{u}_{r,k}^2+\hat{v}_{r,k}^2}\right).\end{aligned} \tag{24}$$

Table I. Simulation parameters

Parameter	Value
Number of transmitting elements M	5
Number of RIS elements Q	5
Number of receiving elements N	16
Number of targets K	3
Number of snapshots L	2000
Transmission angles $(\theta_{t,k}, \varphi_{t,k})$	(45,25),(56,26),(71,48)
Arrival angles $(\theta_{r,k}, \varphi_{r,k})$	(50,21),(71,20),(80,68)
Arrival angles $(\theta_{r,k}, \varphi_{r,k})$	(50,21),(71,20),(80,68)
Polarization parameters (γ_k, η_k)	(30,20),(60,50),(42,68)
Coupling strength c_0	0.8
Coupling attenuation facto β	10
Scaling factor α	(0.8,1.2)
Phase offset ϕ_0	0
Number of Monte Carlo trials M_d	500

3.3. Polarization Parameters Estimation

The matrix $\hat{\mathbf{Q}}_r$ can be reconstructed from the estimated $\mathbf{H}_r$ according to (8). Denoting the k-th column of the matrix as $\hat{\mathbf{B}}_k$, which can be computed as follows:

$$\hat{\mathbf{B}}_k = \frac{(diag(\mathbf{i}_{N-2P,1}) \otimes \mathbf{I}_6)\hat{\mathbf{G}}_2'}{[\mathbf{H}_r]_{1,k}}, \quad k = 1, \ldots, K \tag{25}$$

where $[\mathbf{H}_r]_{1,k}$ denotes the element in the first row and k-th column of a matrix. And, the polarization-dependent vector can be calculated as

$$\hat{\mathbf{g}}_k = \left(\hat{\mathbf{V}}^{\mathrm{H}}\hat{\mathbf{V}}\right)^{\dagger}\hat{\mathbf{V}}^{\mathrm{H}}\hat{\mathbf{b}}_k. \tag{26}$$

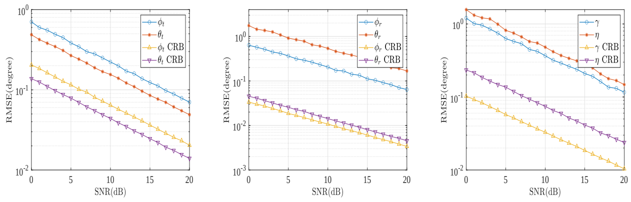
4. SIMULATION RESULTS

In this section, we verify the reliability and anti-coupling capability of the proposed method through simulations. The root mean square error (RMSE) is adopted as the performance metric, and its calculation is given by

$$\mathrm{RMSE} = \sqrt{\frac{1}{M_d K}\sum_{k=1}^{K}\sum_{i=1}^{M_d}\left(\hat{\mu}_{i,k} - \mu_k\right)^2} \tag{27}$$

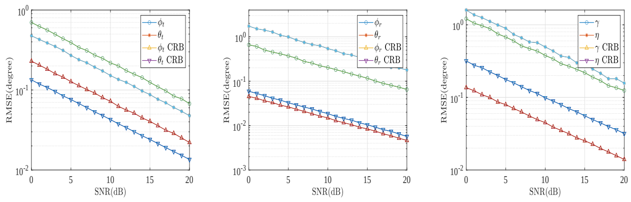
where M_d is the number of Monte Carlo trials, $\hat{\mu}_{i,k}$ is the estimated value of the k-th target parameter in the i-th Monte Carlo trial, and μ_k is the true value of the k-th target parameter.

To investigate the impact of signal-to-noise ratio (SNR) on RMSE performance, the SNR is varied from 0 dB to 20 dB, while the remaining parameters are set as shown in Table I. Figs 2 and 3 respectively present a comparison of the RMSE for angle and polarization parameter estimation, along with the corresponding Cramér-Rao bound (CRB), versus SNR under extended and compressed configurations. Overall, the RMSE for all parameters decreases as SNR increases and gradually approaches the CRB. It can be observed that the



(a) DOD parameters (b) DOA parameters (c) Polarization parameters

Fig. 2. RMSE of parameter estimation versus SNR under the extended configuration.



(a) DOD parameters (b) DOA parameters (c) Polarization parameters

Fig. 3. RMSE of parameter estimation versus SNR under the compressed configuration.

performance curves of the compressed configuration show no significant difference from those of the extended configuration. The performance gap caused by residual mutual coupling is confined to a very small range, and no pronounced performance degradation commonly seen in traditional dense arrays occurs. The presented simulation results demonstrate that, by employing innovative mutual coupling compensation and joint parameter estimation algorithms, the proposed system can achieve high-precision parameter estimation performance in the dense-element compressed configuration that is highly comparable to that of the sparse-element extended configuration. This validates the effective suppression capability of the proposed algorithm against mutual coupling effects.

5. CONCLUSION

To address the challenge of NLOS far-field target localization, this paper innovatively proposes a reconfigurable intelligent surface RIS-aided SEFVA MIMO radar system. By introducing a scalable array structure, intelligent switching between compressed and extended modes is achieved. Combined with techniques such as PARAFAC tensor decomposition and rotational invariance, high-precision joint estimation of 2-D DODs, 2-D DOAs and polarization parameters for multiple far-field targets is realized. Simulation results demonstrate the effectiveness of the proposed method.

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